13: Diagnostic plots for MLR

Stat 230 - Spring 2026

```
library(ggformula)
library(broom)
library(ggResidpanel)
```

1 Data and model

A hospital surgical unit was interested in predicting the survival in patients undergoing a particular type of liver operation. A random selection of 54 patients was selected for analysis. For each patient, the following information was extracted from the pre-operation evaluation and used by the research team to predict survival time (in days):

- `blood_clot_score`: a score based on blood clotting tests
- `prognostic_index`: a prognostic index based on various health factors
- `enzyme_test`: a score based on liver enzyme tests
- `alcohol_use_heavy`: an indicator variable for heavy alcohol use (1 = heavy use, 0 = not heavy use)

The research team determined that the following multiple regression model is a reasonable starting point:

$$\begin{aligned}\text{survival time} = \beta_0 + \beta_1(\text{blood clot score}) + \beta_2(\text{prognostic index}) + \beta_3(\text{enzyme test}) \\ + \beta_4(\text{heavy alcohol use}) + \epsilon\end{aligned}$$

This model is fit using the below code:

```
surgical_data <- read.csv("https://aluby.github.io/stat230/data/surgical_example_data

surgical_lm <- lm(survival_time ~ blood_clot_score + prognostic_index + enzyme_test +
```

2 MLR Conditions

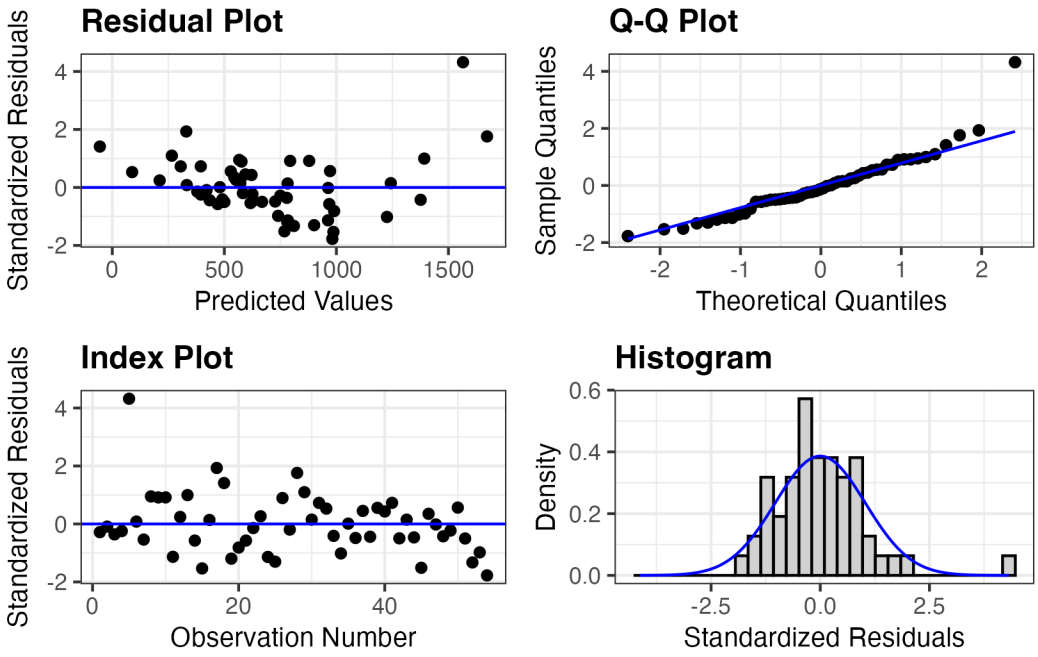
2.1 Recap

- **L**inearity
  - Can the relationship between response Y and predictors X be adequately described with the MLR model?
  - Check using (1) residual plot against fitted values, (2) residual plot against each predictor
  - Any violations should be addressed before conducting inference or making conclusions
- **I**ndependence
  - Are the error terms independent?
  - Check using the data collection process (time/space ordering) or plot of residuals against index
  - Important for conducting inference
- **N**ormal residuals
  - Are the error terms normally distributed?
  - Check using QQplot/histogram
  - Simulation-based inference (eg bootstrap) does not assume shape of residuals. If our sample size is "large" (greater than ≈ 30), the CLT means that inferential methods based on the t distribution are robust to the normality assumption (eg we still get valid results). This is a less important condition.
- **E**qual variance
  - Is σ_ϵ the same for all data points?
  - Check with residuals against fitted. Looking for equal spread above and below the line
  - Violations in the equal variance condition could make inferences based on the t-distribution unreliable. Before conducting inference, we must either address this issue or use a simulation-based approach

2.2 Checking conditions

When we have multiple predictors, we can still use residual plots to assess the conditions of the linear regression model. You can still use the `resid_panel(model, type = "standardized")` function in the *ggResidpanel* package to create a panel of residual plots that includes a plot of the residuals vs. fitted values, a Q-Q plot of the residuals, a histogram of the residuals, and an index plot of the residuals.

```
library(ggResidpanel)
resid_panel(surgical_lm, type = "standardized")
```



We should also check the relationship of the standardized residuals against each of the predictor variables. To do, I recommend using the `augment()` function from the {broom} package to create the augmented dataset, and then making the scatterplots with the graphing code of your choice.

Q1: Use the augmented dataset below to make scatterplots of the standardized residuals against (1) `blood_clot_score`, (2) `prognostic_index`, (3) `enzyme_test`, (4) `alcohol_use_heavy`

Hint: for the 0/1 indicator, a boxplot is easier to read than a scatterplot.

```
surgical_aug <- augment(surgical_lm)

# your code here
```

Q2: Based on the residual plots, do you think the current multiple linear regression model is appropriate for this data? Why or why not?

Q3: Regardless of your answer, apply a log transformation to the response variable and refit the model using the code below. Then, create the same residual plots for the new model. Do you think the log transformation improved the model fit? Why or why not?

```
log_surgical_lm <- lm(log(survival_time) ~ blood_clot_score + prognostic_index + enzy

# Put your residual plot code here
```

3 MLR Influence analysis

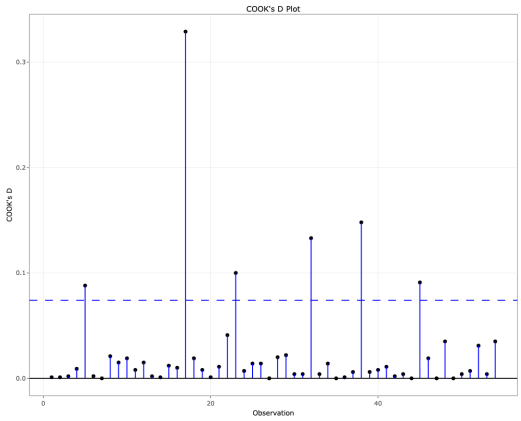
Let's continue working with the log-transformed model.

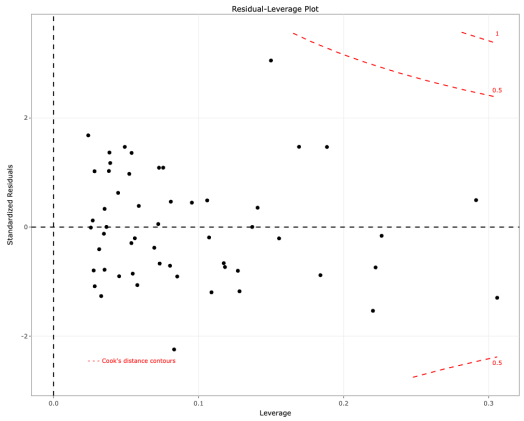
Q4: Use `resid_panel` to make plots of (a) the standardized residual vs fitted values, (b) the residual vs leverage plot, (c) cook's distance plot. Does anything stand out?

```
# resid_panel(model, plots = c("resid", "lev", "cookd"), type = "standardized")
```

Q5: The `resid_interact` function makes an interactive version of the plots which allow you to hover on the points to identify which point (index) they correspond to. What are the observation number (index) of any points you noticed above?

```
resid_interact(log_surgical_lm, plots = c("cookd", "lev"))
```





Q6: Do you think any of these observations are truly influential? Why or why not? It may be useful to look back through your notes to review the definitions of these influence measures and the suggested cutoffs. (For this model, $n = 54$ and $p = 4$.)

Q7: Refit the log-transformed model without the most influential observation you identified in Q6, and compare the coefficient estimates to the original fit. Following the flowchart from today's notes, what would you do with this case?

```
# Refit the log-transformed model, leaving out one row of the data
log_surgical_lm2 <- lm(log(survival_time) ~ blood_clot_score + prognostic_index + enz
                       data = surgical_data[-___, ])
```

3.1 Function quick reference

Function	Purpose
`augment()`	Add fitted values, residuals, leverage (`.hat`), standardized residuals (`.std.resid`), and Cook's distance (`.cooksd`) to the data
`resid_panel()`	Make a panel of residual plots (use `type = "standardized"`)
`resid_interact()`	Make interactive residual plots to identify individual observations